

CNN

Question 4

Alice is a BRACU student and she is taking CSE428 this semester. For her final project, she is trying to implement a CNN architecture for a classification task that comprises of the following layers:

Layer	Input Dimensions	Filter Size	#Filters or, #Neurons	Padding	Output Dimensions	#Params
Conv1	128 * 128 * 3	7*7	8	2		
MaxPool1		2*2	—	0		
Conv2		5*5	16	2		
MaxPool2		2*2	—	0		
Conv3		3*3	32	0		
AvgPool3		4*4	—	0		
Flatten		—	—	—		
FC		—	256	—		
FC		—	128	—		
FC (Output)		—	4	—		

In the table above, *Conv-X* denotes a Convolutional layer, *Pool-X* denotes a Pooling layer and *FC* denotes a Fully Connected layer.

- Determine the number of classes and the activation function used in the final layer.
- Calculate the input and output dimensions for each of the layers. (Complete the 2nd and 6th columns of the table).
- Calculate the number of Parameters for each of the layers. (Complete the last column of the table).
- Repeat Question b considering a Mini-Batch size of 32 instead of individual inputs.
- Suppose, Alice used Batch Normalization layers after each Convolutional and Fully Connected layer. Would it Change the total number of trainable parameters? If Yes, then, by how much?

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Layer	Input Dimensions	Filter Size	# Filters or Neurons	Padding	Output Dimensions	Parameters
Conv1	$128 * 128 * 3$	$7 * 7$	8	2	$126 * 126 * 8$	$((7 * 7 * 3) + 1) * 8 = 1184$
MaxPool1	$126 * 126 * 8$	$2 * 2$	—	0	$63 * 63 * 8$	0
Conv2	$63 * 63 * 8$	$5 * 5$	16	2	$63 * 63 * 16$	$((5 * 5 * 8) + 1) * 16 = 3216$
MaxPool2	$63 * 63 * 16$	$2 * 2$	—	0	$31 * 31 * 16$ (floor value)	0
Conv3	$31 * 31 * 16$	$3 * 3$	32	0	$29 * 29 * 32$	$((3 * 3 * 16) + 1) * 32 = 4640$
AvgPool3	$29 * 29 * 32$	$4 * 4$	—	0	$7 * 7 * 32$ ($\lfloor 29/4 \rfloor$)	0
Flatten	$7 * 7 * 32$	—	—	—	$7 * 7 * 32 = 1568$	0
FC	1568	—	256	—	256	$(1568 * 256) + 256 = 401664$
FC	256	—	128	—	128	$(256 * 128) + 128 = 32896$
FC (Output)	128	—	4 multiclass classification	—	4	$(128 * 4) + 4 = 516$

Last activation $\rightarrow$ Softmax (multiclass classification)

In that case, Output dimension = Filters

Binary classification (filter = 2) $\rightarrow$ Sigmoid

In that case, Output dimension = 1

Convolutional Layer

$$W_{out} = \frac{W_{in} - F + 2 * P}{\# \text{ stride}} + 1$$

$$= \frac{128 - 7 + 2 * 2}{1} + 1$$

$$= 126$$

$$H_{out} = \frac{H_{in} - F + 2 * P}{\# \text{ stride}} + 1$$

$$= \frac{128 - 7 + 2 * 2}{1} + 1$$

$$= 126$$

C_{out} = Number of Filters applied on the input images.

$$= 8$$

Input Dimensions

$$\rightarrow [W_{in}, H_{in}, C_{in}]$$

For Batch Size $[N, W_{in}, H_{in}, C_{in}]$

Output Dimensions

$$\rightarrow [W_{out}, H_{out}, C_{out}]$$

For Batch Size $[N, W_{out}, H_{out}, C_{out}]$

F = Kernel Dimension
(F for an $F * F$ kernel)

P = Padding Length

$\# \text{ Stride} = \text{strides}$

$$= 1$$

Pooling Layer

$$W_{out} = \frac{W_{in} + 2 * P}{F}$$

$$= \frac{126 + 2 * 0}{2} = 63$$

$$H_{out} = \frac{H_{in} + 2 * P}{F}$$

$$= \frac{126 + 2 * 0}{2} = 63$$

$$C_{out} = C_{in} = 8 \text{ (no change)}$$

Flattening

Input Dimensions $\rightarrow [W_{in}, H_{in}, C_{in}]$

Then, Output Dimensions $\rightarrow W_{in} \times H_{in} \times C_{in} \rightarrow \# \text{ output units}$

Parameter Calculation

Convolutional Layer:

$$\text{Parameters} = \{(k_w \times k_h \times c_{in}) + 1\} \times C_{out}$$

where:

k_w = kernel width

k_h = kernel height

c_{in} = number of input channels

C_{out} = number of filters (output channels)

The "+1" accounts for the bias term for each filter.

Fully connected Layer:

$$\text{Parameters} = (\# \text{ input units} \times \# \text{ output units}) + \# \text{ output units}$$

$$= (128 \times 64) + 64 = 8256$$

→ 128 input layer units

→ 64 output layer units

Pooling Layer:

$$\text{Parameter} = 0$$

Batch Normalization layer:

$$\text{Parameters} = 2 \times C_{out}$$

If a batch normalization layer is applied after a fully connected layer,

$$\text{Parameters} = 2 \times \# \text{ output units}$$

a)

Number of classes: There are 4 classes, as indicated by the 4 neurons in the FC (output) layer.

Activation Function: For a multi-class classification task with more than 2 classes, the softmax activation function is used in the final layer.

d)

Layer	Input Dimensions	Output Dimensions
conv 1	$32 \times 128 \times 128 \times 3$	$32 \times 126 \times 126 \times 8$
Maxpool 1	$32 \times 126 \times 126 \times 8$	$32 \times 63 \times 63 \times 8$
conv 2	$32 \times 63 \times 63 \times 8$	$32 \times 63 \times 63 \times 16$
Maxpool 2	$32 \times 63 \times 63 \times 16$	$32 \times 31 \times 31 \times 16$
conv 3	$32 \times 31 \times 31 \times 16$	$32 \times 29 \times 29 \times 32$
Avg pool 3	$32 \times 29 \times 29 \times 32$	$32 \times 7 \times 7 \times 32$
Flatten	$32 \times 7 \times 7 \times 32$	$32 \times 7 \times 7 \times 32$
FC	1568×32	32×25
FC	32×256	32×128
FC (output)	32×128	32×4

e) Yes, this changes the number of trainable parameters.

A standard Batch Normalization layer has 4 parameters per channel/neuron: two trainable (γ and β) and two non-trainable (moving mean and variance).

Calculations for Trainable Parameters:

$$\text{Parameter} = 2 \times \text{Count}$$

↳ number of filters

$$\text{After Conv1} : 2 \times 8 = 16$$

$$\text{After Conv2} : 2 \times 16 = 32$$

$$\text{After Conv3} : 2 \times 32 = 64$$

$$\text{After FC (256)} : 2 \times 256 = 512$$

$$\text{After FC (128)} : 2 \times 128 = 256$$

$$\text{Total increase} : 16 + 32 + 64 + 512 + 256 = 880$$

additional trainable parameters.